EUROPEAN JOURNAL OF PSYCHOTRAUMATOLOGY
2025, VOL. 16, NO. 1, 2541473
https://doi.org/10.1080/20008066.2025.2541473

EDITORIAL

OPEN ACCESS Check for updates

Biological mechanisms underlying adverse mental health outcome after trauma

Mirjam van Zuiden [a,b] and Miranda Olff [b,c]

[a]Department of Clinical Psychology, Utrecht University, Utrecht, the Netherlands; [b]Department of Psychiatry, Amsterdam University Medical Centers, University of Amsterdam, Amsterdam Public Health, Amsterdam Neuroscience, Amsterdam, the Netherlands; [c]ARQ National Psychotrauma Centre, Diemen, the Netherlands

ABSTRACT

By now it is well known that trauma-related psychiatric disorders are associated with a broad range of biological dysregulations. This special issue focused on research aiming to advance our understanding of the biological mechanisms underlying adverse mental health outcome after trauma, in particular the development of trauma-related disorders and differences in individual susceptibility (i.e. risk or resilience) to their development. We included 51 papers published in EJPT between 2021 and 2023, either directly submitted to the special issue or whose focus clearly fitted the aim of the special issue. Included papers consisted of 43 original research papers and 8 review articles, addressing a wide range of biological processes, ranging from brain and central nervous system functioning, to cardiovascular and autonomic nervous system functioning; endocrinology; (epi)genetics; metabolism; inflammation; sleep and pain. Collectively, the included articles showcase that significant steps are taken in advancing our understanding of the biological mechanisms underlying development of and susceptibility to trauma-related disorders, as well their recovery. Recommendations for future research are provided with regard to study design; assessment of trauma exposure characteristics; sex and/or gender (minority)-related factors; geographic, developmental/lifespan and transgenerational aspects; and the temporal dynamics of biological processes in relation to the stage of the disorder under investigation. Together, these articles will hopefully further facilitate the development of more effective preventive, early and curative intervention strategies, as well as more effective timely and personalised allocation towards these interventions.

Mecanismos biológicos subyacentes a las consecuencias adversas para la salud mental tras un trauma

Hasta el momento es bien sabido que los trastornos psiquiátricos relacionados con trauma se asocian a una amplia gama de desregulaciones biológicas. Este número especial se centró en investigación destinada a profundizar nuestra comprensión de los mecanismos biológicos que subyacen a las consecuencias adversas para la salud mental tras un trauma, en particular el desarrollo de los trastornos relacionados con trauma y las diferencias en la susceptibilidad individual (es decir, riesgo o resiliencia) d su desarrollo. Se incluyeron 51 artículos publicados en la EJPT entre 2021-2023, ya sea enviados directamente al número especial o cuyo foco se ajustaba claramente al objetivo del mismo. Estos incluyeron 43 artículos de investigación originales y 8 artículos de revisión, que abordan una amplia gama de procesos biológicos, variando desde el funcionamiento cerebral y sistema nervioso central, al funcionamiento cardiovascular y sistema nervioso autónomo; endocrinología; (epi)genética; metabolismo; inflamación; sueño y dolor. Colectivamente, los artículos incorporados muestran que se están dando pasos importantes para avanzar en nuestra comprensión de los mecanismos biológicos subyacentes al desarrollo y la susceptibilidad a los trastornos relacionados con trauma, así como su recuperación. Se dieron recomendaciones para futuras investigaciones con respecto a diseño de estudio; evaluación de las características de exposición al trauma; factores relacionados con (minorías de) sexo y/o género; aspectos geográficos, del desarrollo/a lo largo de la vida y transgeneracionales; y las dinámicas temporales de los procesos biológicos en relación con la etapa del trastorno bajo investigación. Se espera que, en conjunto, estos artículos faciliten aún más el desarrollo de estrategias de intervención preventivas, tempranas y curativas más eficaces, así como una asignación mas eficaz, oportuna y personalizada a estas intervenciones.

KEYWORDS
PTSD; neuroscience; brain; nervous system; metabolism; immunology; metabolism; endocrinology; cardiovascular; (epi)genetics

PALABRAS CLAVE
TEPT; neurociencias, cerebro, sistema nervioso; metabolismo; inmunología; metabolismo; endocrinología; cardiovascular; (epi)genética

HIGHLIGHTS

- This special issue based on contains 51 papers showcases incremental knowledge about biological mechanisms of trauma-related disorder development, susceptibility and recovery.
- More research is needed with more varied and understudied populations in terms of geographic location, developmental and aging perspectives.
- Incorporation of trauma exposure characteristics and sex and/or gender effects into descriptive and analytic approaches is needed.
- More precise information is needed on timing of biological measures in relation to the trauma exposure and the stage of the disorder.

CONTACT Mirjam van Zuiden m.vanzuiden@uu.nl Department of Clinical Psychology, Utrecht University, Utrecht, the Netherlands

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group
This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (http://creativecommons.org/licenses/by-nc/4.0/), which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

1. Introduction

Trauma-related disorders including posttraumatic stress disorder (PTSD) are associated with dysregulations in a wide range of biological processes. These processes include but are not confined to the central nervous system, autonomic nervous system; endocrine system; immune system and metabolic system. An extensive review of these systems and their associations with trauma-related is beyond the scope of this editorial, but interested readers are referred to the following recent overview paper (Shalev et al., 2024). Initial biological studies in our field mainly focused on the associations between biological measures and the presence or maintenance of trauma-related disorders. However, more recent studies provide accumulating evidence that biological dysregulations should not merely be considered as an acquired consequence of exposure to traumatic events or trauma-related disorders. In contrast, several biological processes seem to be associated with emotional and cognitive trauma processing and symptom onset, and are increasingly thought to contribute to individual differential susceptibility (i.e. risk or resilience) to the development of trauma-related disorders.

PTSD and other-trauma related disorders are highly prevalent and once their symptoms have established the odds for a chronic adverse course are considerable, even when evidence-based treatment is provided (Kessler et al., 2017; Semmlinger et al., 2024). Thus, further elucidation of the biological processes associated with the development of and susceptibility to trauma-related disorders is important to provide further avenues for much warranted preventive and early interventions to ameliorate long-term adverse mental health consequences of trauma. Therefore, this special issue focused on research aiming to advance our understanding of these biological processes. We specifically called for studies with adequate designs and setups to investigate biological processes in relation to relevant clinical, cognitive or behavioural aspects of trauma exposure and development of and susceptibility to trauma-related disorders, including prospective and longitudinal cohort studies, but also experimental, preclinical and computational studies. We specifically invited studies addressing: acute and long-term biological effects of traumatic stress; biological factors associated with risk and resilience to trauma-related disorders; biological processes of symptom onset and development after trauma; biological information to forecast mental health outcome after trauma; mediators and moderators of biological mechanisms of trauma-related disorders; intervention studies targeting or assessing biological processes associated with development of trauma-related disorders; impact of sex as a biological variable (SABV) on mental health outcome after trauma; other biological papers relevant to the field of psychotraumatology

2. In this issue

In this special issue we collected 51 papers accepted for publication in EJPT between 2021 and 2023.

The papers eventually selected for the special issue were either submitted specifically in response to the abovementioned call for papers until the end of 2022, or selected from articles accepted for publication in 2023 based on their clear biological focus. Forty-three of the included articles consisted of original research papers, with approximately 60% ($n = 26$) describing results of studies with cross-sectional study designs. The other original research papers presented results from studies with a prospective ($n = 5$); longitudinal ($n = 4$), experimental (i.e. analogue trauma) ($n = 4$) or clinical trial ($n = 4$) design. Within these original research articles, the overlarge majority ($n = 32$, 74.4%) focused on biological correlates of trauma-related psychiatric disorders or characteristic symptoms thereof, predominantly posttraumatic stress disorder (PTSD). The remaining 11 original research papers concerned investigations of the associations between trauma exposure and biological processes, without considering its potential clinical consequences or in some instances excluding individuals with trauma-related disorders. Additionally, the special issue collection contains two narrative reviews, two systematic reviews with meta-analytic components, and four systematic reviews. Aligning with the trend observed in the original research papers, six of these focused specifically on PTSD or its symptoms, whilst the other two focused on the consequences of trauma exposure without a particular focus on its clinical consequences. All details of the papers are provided in the enclosed tables.

2.1. Biological processes investigated

The biological processes investigated within the articles included in this special issue can be broadly clustered as follows: brain/central nervous system; cardiovascular/autonomic nervous system; endocrinology; (epi)genetics; and a miscellaneous group including sleep, pain, metabolism and inflammation. Below the articles are clustered based on their primary biological focus, but please note that several articles focused on more than one of these processes.

2.1.1. Brain / central nervous system

Brain and/or CNS-related research was the most prevalent primary focus of the articles accepted into the special issue, with 21 articles with this particular focus. Of these, 16 presented results from original research: One prospective study (Powers et al., 2022), one (Lan et al.) longitudinal study (Lan et al., 2023),

and 14 studies with a cross-sectional design (Glazebrook et al., 2023; He et al., 2023; Hilberdink et al., 2021; Jin et al., 2023; Liddell et al., 2023; Rosada et al., 2021; Schellhaas et al., 2022; Sokołowski et al., 2022; Tamman et al., 2023; van Sleeuwen et al., 2023; Vicario et al., 2022; Wermuth et al., 2021; Xiao et al., 2023; Zhao et al., 2021) Of the remaining 5 articles not presenting original research, four consisted of systematic reviews (one including a meta-analytic investigation; (Askovic et al., 2023); purely systematic (Manthey et al., 2021; Nicholson et al., 2022; Susanty et al., 2022) and one consisted of a narrative review (Fine et al., 2023). The exact biological processes under investigation as well as the applied assessment methods for each of the 21 articles are depicted in Table 1.

2.1.2. Cardiovascular / autonomic nervous system

Ten of the articles within the special issue had a primary focus on cardiovascular and/or autonomic nervous system-related research. Of these, nine articles presented results from original research: 5 were cross-sectional studies (Bussone et al., 2023; Grasser et al., 2022; Krause-Utz et al., 2022; Noble et al., 2023; Schmitz et al., 2021), 2 were experimental studies (Beutler & Croy, 2023; Danböck et al., 2021) and 2 described results from clinical trials (Magnin et al., 2023; Roullet et al., 2022). The final article consisted of a systematic review (Beutler et al. 2022). The exact biological processes under investigation as well as the applied assessment methods for each of these 10 articles are depicted in Table 2.

2.1.3. Endocrinology

Eight of the articles within the special issue described original research results with a primary focus on endocrinology. These articles consisted of one prospective study (Yrondi et al., 2022), 3 cross-sectional studies (Krinke et al., 2022; Lee et al., 2022; Maslahati et al., 2023; Salvat-Pujol et al., 2021; Wadji et al., 2021), 2 experimental studies (Krinke et al., 2022; Maslahati et al., 2023), and 1 clinical trial (Engel et al., 2021). The final article consisted of a systematic review and meta-analysis (Engel et al., 2023). The exact biological processes under investigation as well as the applied assessment methods for each of these eight articles are depicted in Table 3.

2.1.4. (Epi)genetics

Six articles within the special issue primarily focused on epigenetic or genetic investigations. These studies all presented results from original research, with 2 articles described results from cross-sectional studies (Cusack et al., 2023; Mattonet et al., 2023); 2 articles describing results from longitudinal studies (dos Santos Oliveira et al., 2023; Elam et al., 2022) and 2 describing results from prospective studies (Gultig et al., 2023; Vuong et al., 2022). The exact biological processes under investigation as well as the applied assessment methods for each of these six articles are depicted in Table 4.

2.1.5. Miscellaneous

The remaining 6 articles contain a mixture of primary foci. Two original research articles focus on sleep, one with a cross-sectional (Ansbjerg et al., 2023) and one with a longitudinal design (Schenker et al., 2023). Three papers describe original research related to inflammatory and/or metabolic processes, including one cross-sectional study (Kurkinen et al., 2023), one prospective study (Baranyi et al., 2021) and one clinical trial (Maffioletti et al., 2021). The final article presents a narrative review encompassing a multitude of biological processes in relation to childhood trauma and physical pain (Campbell, 2022). The exact biological processes under investigation as well as the applied assessment methods for each of these six articles are depicted in Table 5.

3. Highlights and limitations

Collectively, the included articles in our special issue investigated a wide range of biological processes in relation to trauma and trauma-related disorders, ranging from brain and central nervous system functioning; to cardiovascular and autonomic nervous system functioning; to endocrinology; (epi)genetics; metabolism; inflammation; sleep and pain. The most common research design of the included articles consisted of cross-sectional comparisons of biological measures between participants with or without trauma-related psychiatric disorders (mostly PTSD), or with or without previous trauma exposure. Many of these articles additionally incorporated other trauma-relevant psychological, cognitive and behavioural processes previously associated with mental health outcome following trauma, with the aim to shed light on potential mediators and moderators of the associations between the investigated biological processes and trauma exposure and its related disorders. Beyond the more traditional cross-sectional studies, we also included multiple articles with prospective, longitudinal, experimental designs as well as clinical treatment RCTs. These latter designs are better suited to investigate the temporal courses of the associations of the investigated biological processes with trauma exposure, and with the development, maintenance as well as the recovery of trauma-related disorders. Therefore, we recommend future studies aiming to obtain more mechanistic insights to prioritise such designs over cross-sectional designs.

We were happy to see that the participant samples investigated in the included articles were quite diverse

Table 1. Overview of exact biological processes under investigation as well as the applied assessment methods for each of the 21 articles with a primary focus on brain/central nervous system.

Authors	Year	Global recruitment location	Study design	Trauma Type	Primary disorder	Intervention type	Biological processes	Biological assessment	Other constructs investigated	Time since trauma at assessment	Stage of disorder at assessment	Developmental period at assessment
Askovic et al.	2023	various, not specified	systematic review & meta-analysis	various	PTSD	electroencephalographic neurofeedback	neural and electrocortical activity	fMRI: various and EEG: various	na	not specified	various	adulthood
Glazebrook et al.	2023	Australia	cross-sectional	various	PTSD	na	electrocortical activity	EEG	resilience; posttraumatic growth	not specified	not specified	adulthood
He et al.	2023	China	cross-sectional	childhood trauma	anxiety; depression	na	white matter integrity	diffusion tensor imaging	na	long-term	not specified	adulthood
Fine et al.	2023	na	narrative review	not specified	PTSD	na	neural connectivity and reactivity	na	guilt, shame	not specified	not specified	not specified
Hilberdink et al.	2021	Netherlands	cross-sectional	childhood trauma	PTSD	na	neural connectivity, autonomic reactivity, HPA axis reactivity	fMRI: functional connectivity; cortisol, cortisol awakening response, heart rate	social-evaluative stress	not specified	not specified	adolescence
Jin et al.	2023	China	cross-sectional	childhood trauma	na	na	neural connectivity	fMRI: functional connectivity	na	long-term	na	young adulthood
Lan et al.	2023	China	longitudinal	child loss	PTSD; anxiety; depression	na	brain structure	MRI: cortical thickness and cortical surface area	na	long-term	subclinical	adulthood
Liddell et al.	2023	Australia, refugees from various other countries (majority Middle-East)	cross-sectional	refugee/war	na	na	neural connectivity	fMRI: functional connectivity	na	ongoing	na	adulthood
Manthey et al.	2021	various, not specified	systematic review	not specified	PTSD	psychological treatment for PTSD (various types)	neural connectivity and reactivity; brain structure	fMRI: various; MRI: various	na	not specified	treatment seeking	adulthood
Nicholson et al.	2022	various settings in Europe, Northern America and Asia	systematic review	sexual minority stress	na	na	neural connectivity and reactivity; brain structure	fMRI: various; MRI: various	na	not specified	na	young adulthood
Powers et al.	2022	USA	prospective	accidental injury	PTSD	na	neural activity	fMRI: task-based	behavioural response inhibition	early	development/ susceptibility	adulthood
Rosada et al.	2021	Germany	cross-sectional	childhood trauma	PTSD; Borderline personality disorder	na	brain structure	MRI: brain volume	na	long-term	not specified	adulthood
Schellhaas et al.	2022	Germany	cross-sectional	childhood trauma	na	na	electrocortical activity	EEG	item/source memory; visual working memory; threat/safety signals	long-term	na	adulthood
Sokolowski et al.	2022	Poland	cross-sectional	childhood trauma	na	na	neural connectivity	fMRI: functional connectivity	rumination	long-term	na	adulthood
Susanty et al.	2022	USA, Netherlands, Germany, Iran	systematic review	various	PTSD	psychological treatment for PTSD (various types)	neurocognition	na	attention; memory; information processing; executive function	not specified	treatment seeking	adulthood
Tamman et al.	2023	USA	cross-sectional	childhood trauma	PTSD	na			na	long-term	not specified	adulthood

							glutamatergic synaptic strength	magnetic resonance spectroscopy				
van Sleeuwen et al.	2023	Netherlands	cross-sectional	police work-related trauma; childhood trauma	PTSD	na	neural activity; brain structure; neuropeptides/ hormones	fMRI: task-based; MRI: voxel-based morphometry; oxytocin	alexithymia	not specified	not specified	adulthood
Vicario et al.	2022	Australia	cross-sectional	not specified	PTSD	na	brain structure	MRI: grey matter voxel-based morphometry	time processing	not specified	not specified	adulthood
Wermuth et al.	2021	Germany	cross-sectional	interpersonal violence	na	na	brain structure	MRI: white matter voxel-based morphometry	emotion regulation	early	na	adulthood
Xiao et al.	2023	China	cross-sectional	na	borderline personality disorder	na	neural connectivity; brain structure	fMRI: functional connectivity; MRI: cortical thickness	emotion regulation	na	chronic	adolescence
Zhao et al.	2021	China	cross-sectional	childhood trauma	na	na	neural connectivity	fMRI: functional connectivity	na	longterm	na	adulthood

Table 2. Overview of exact biological processes under investigation as well as the applied assessment methods for each of the 10 articles with a primary focus on cardiovascular/autonomous nervous system.

Authors	Year	Global recruitment location	Study design	Trauma type	Primary disorder	Intervention type	Biological processes	Biological assessment	Other constructs investigated	Time since trauma at assessment	Stage of disorder at assessment	Developmental period at assessment
Beutler and Croy	2023	Germany	experimental	trauma film	PTSD (intrusions)	na	autonomic reactivity	heart rate; heart rate variability; body sway; trembling frequency	intrusions	peri; acute	development/ susceptibility	adulthood
Beutler et al.	2022	not specified	systematic review	various	PTSD (dissociation)	na	autonomic reactivity	heart rate, heart rate variability, blood pressure, skin conductance	dissociation	various	various	adulthood
Bussone et al.	2023	Italy	cross-sectional	childhood trauma	na	na	autonomic activation	heart rate variability	non verbal communication	longterm	na	adulthood
Danböck et al.	2021	Austria	experimental	trauma film	PTSD (dissociation, intrusions)	na	autonomic activation	heart rate, respiratory sinus arrhytmia, skin conductance, facial electromyography reactivity	dissociation; intrusive memories	peri; acute	development/ susceptibility	adulthood
Krause-Utz et al.	2022	Germany	cross-sectional	childhood trauma	Borderline personality disorder	na	autonomic reactivity	heart rate variability	emotional working memory	not specified	not specified	adulthood
Magnin et al.	2023	France	clinical trial	various	Acute Stress Disorder/ PTSD	prazosin (alpha 1 adrenergic receptor blocker)	noradrenergic receptor signalling	na	na	acute	acute; early	adulthood
Noble et al.	2023	USA	cross-sectional	lifetime, nos	PTSD; MDD	na	autonomic activation	blood pressure, heart rate, heart rate variability	na	not specified	not specified	adulthood
Roullet et al.	2022	France	clinical trial	lifetime, nos	PTSD (with comorbid dissociation)	propanalol (beta-adrenergic receptor blocker)	noradrenergic receptor signalling	na	memory reactivation	longterm	chronic	adulthood
Grasser et al.	2022	USA; participants resettled refugees from Syria; Iraq	cross-sectional	refugee/war	PTSD	na	autonomic reactivity	skin conductance	na	not specified	not specified	childhood/ adolescence
Schmitz et al.	2021	Germany	cross-sectional	various, including childhood trauma	PTSD	na	autonomic activation	ECG: heartbeat evoked potentials	dissociation, interoception	not specified	not specified	adulthood

Table 3. Overview of exact biological processes under investigation as well as the applied assessment methods for each of the 8 articles with a primary focus on endocrinology.

Authors	Year	Global recruitment location	Study design	Trauma type	Primary disorder	Intervention type	Biological processes	Biological assessment	Other constructs investigated	Time since trauma at assessment	Stage of disorder at assessment	Developmental period at assessment
Engel et al.	2021	Germany	clinical trial	military	PTSD	internet-based TF-CBT	neuropeptides/ hormones	oxytocin, vasopressin	na	not specified	treatment seeking	adulthood
Engel et al.	2023	not specified?	systematic review & meta-analysis	trauma film	PTSD (intrusions)	na	HPA axis reactivity	cortisol	intrusions	peri; acute	development / susceptibility	adulthood
Krinke et al.	2022	Switzerland	experimental	trauma film	PTSD (intrusions)	na	sex hormones; HPA axis	progesteron, estradiol, testosteron, cortisol	intrusions	peri; acute	development / susceptibility	adulthood
Lee et al.	2022	South Korea	cross-sectional	childhood trauma	not specified	na	HPA axis activity	hair cortisol concentrations	theory of mind	not specified	na	adulthood
Maslahati et al.	2023	Germany	experimental	trauma film	PTSD (intrusions)	na	sex hormones	Oral contraceptive use	intrusions	peri; acute	development / susceptibility	adulthood
Salvat-Pujol et al.	2021	Spain	cross-sectional	childhood	MDD	na	HPA axis reactivity	cortisol following dexamethasone suppresion test	verbal memory, visual memory and executive function/ processing speed	not specified	treatment seeking	adulthood
Wadji et al	2021	Cameroon	cross-sectional	intimate partner violence	na	na	HPA axis reactivity	cortisol awakening response	self-esteem	past 12 months	na	adulthood and childhood - family dyads
Yrondi et al.	2022	France	prospective	accidental injury	PTSD/MDD	na	HPA axis activitiy	morning cortisol	na	early	development / susceptibility	adulthood

Table 4. Overview of exact biological processes under investigation as well as the applied assessment methods for each of the 6 articles with a primary focus on (epi)genetics.

Authors	Year	Global recruitment location	Study design	Trauma type	Primary disorder	Intervention type	Biological processes	Biological assessment	Other constructs investigated	Time since trauma at assessment	Stage of disorder at assessment	Developmental period at assessment
Cusack et al.	2023	US	cross-sectional	lifetime, nos	internalizing symptoms	na	genetic polymorphisms	genome-wide association studies (GWAS), genome-wide complex trait analysis (GCTA), polygenic risk score (PRS) analyses	resilience	na	not specified	adulthood
dos Santos Oliveira et al.	2023	Brazil	longitudinal	community and domestic violence	na	na	epigenetic aging; heart rate variability/vagal activitiy	epigenetic clocks (GrimAge); vagal activity/heart rate variability	na	lifetime; past 5 years	na	adolescence
Elam et al.	2022	US	longitudinal	interparental conflict, childhood trauma	na	family-based divorce related intervention	epigenetic aging	telomere length	na	longterm	na	adulthood and adolescence - family dyads
Gultig et al.	2023	Netherlands	prospective	prenatal famine; childhood trauma (NOS); adulthood trauma (NOS)	PTSD	na	Glucocorticoid receptor genetic polymorphisms	Glucocorticoid receptor single nucleotide polymorphisms	na	na	not specified	late adulthood
Mattonet et al.	2023	Tanzani, refugees from Burundi	cross-sectional	refugee/war	PTSD	na	epigenetic methylation	epigenome-wide association studies (EWAS): methylation in peripheral buccal epithelia cells	na	approximately 2 years since latest conflict; with ongoing chronic stress	not specified	adulthood and childhood/ adolescence - family dyads/ triads
Vuong et al.	2022	South Africa	prospective	sexual violence; childhood trauma	PTSD	na	Adiponectin genetic polymorphisms	Adiponectin single nucleotide polymorphisms	na	na	development / susceptibility	adulthood

Table 5. Overview of exact biological processes under investigation as well as the applied assessment methods for each of the 6 articles with a miscellaneous primary focus.

Authors	Year	Global recruitment location	Study design	Trauma type	Primary disorder	Intervention type	Biological processes	Biological assessment	Other constructs investigated	Time since trauma at assessment	Stage of disorder at assessment	Developmental period at assessment
Baranyi et al.	2021	Austria	prospective	cardiovascular disease	PTSD	na	gut microbial metabolism	Trimethylamine N-oxide (TMAO) serum levels	na	acute; early	development / susceptibility, early chronicity	adulthood
Campbell	2022	na	narrative review	childhood trauma	na	na	various; including neural activity, HPA axis, gene expression, genetic polymorphisms	various	pain	not specified	na	childhood
Maffioletti et al.	2021	Italy	clinical trial	childhood trauma and recent life events	MDD	Trauma-focused psychotherapy	inflammation; micro RNAs	inflammation-related candidate micro RNA expression levels	na	multiple, longterm and recent	treatment resistant	adulthood
Schenker et al.	2023	Australia	longitudinal	not specified	PTSD	na	sleep	objective sleep (actigraphy)	na	not specified	not specified	adulthood
Ansbjerg et al.	2023	Denmark – refugees from Iran/Iraq/ Afghanistan/ Lebanon/Syria/ other	cross-sectional	refugee/war	PTSD	na	sleep	sleep architecture - polysomnography	na	not specified	not specified	adulthood
Kurkinen et al.	2023	Finland	cross-sectional	sexual and physical child abuse	MDD	na	metabolomics	serum metabolite concentration	na	not specified	not specified	adolescence

in terms of demographic and exposure characteristics. In terms of recruitment location, many of the investigated samples extended well beyond the so called WEIRD (Western, Educated, Industrial, Rich, and Democratic) populations, facilitating generalizability of findings to the global population. Furthermore, whilst most investigated samples in this issue consisted of adults, several studies focused specifically on childhood, adolescence or young or late adulthood (for a systematic review on biopsychosocial factors that may influence the onset and persistence of PTSD in children and adolescence also see (Williams et al., 2024). Furthermore, 3 studies assessed child-adult family dyads and even triads (Elam et al., 2022; Mattonet et al., 2023; Wadji et al., 2021). As such, these studies will contribute to elucidating the developmental and even transgenerational aspects of trauma and how these may impact potential adverse mental health outcomes. Future studies would ideally include even more varied and understudied populations in terms of location as well as developmental and aging perspectives.

Participants across articles had been exposed to a wide variety of (potential) traumatic events. This is relevant as trauma characteristics, including the type of trauma exposure, are known to impact the risk for subsequent development of trauma-related disorders including PTSD, as well as their course and phenotypical symptom expression (Brewin et al., 2025). However, in particular for the cross-sectional and clinical intervention studies, trauma type and the broader trauma history were often not specified or taken into account otherwise. Not all included articles in this special issue strictly specified trauma according to the DSM-5 definition, instead investigating populations with more chronic and perhaps more insidious stress exposure (e.g. Elam et al., 2022; Gultig et al., 2023). As there is ongoing debate about the optimal conceptualisation and demarcation of what experiences should be considered traumatic (Marx et al., 2024), such studies will contribute to explorations of potential similarities and differences in the biological correlates between these types of exposures. Therefore, to advance our understanding of if and how these characteristics interact with the biological processes associated with trauma exposure trauma-related adverse biological outcomes, it is recommended that future biological studies incorporate trauma exposure characteristics into their descriptive and analytic approach.

Accumulating findings support that many of biological processes investigated in the included articles are temporally dynamic: their functioning as well as their associations with trauma-related disorders depend on the time since trauma exposure onset and exact stage of the disorder at the time of assessment. These observations fit with the more recent perspective that psychiatric disorders including PTSD and other-trauma related disorders should be approached from a staging perspective, if we are to better understand their etiology and prognosis and improve treatment outcomes (Burback et al., 2024). The timing of assessments relative to exposure and symptom onset is commonly well reported for prospective studies, including those within the special issue. Additionally, the included experimental studies inducing so-called analogue traumatic stress had detailed reporting on the exact timing of their peri- and acute post-'traumatic' assessments (Beutler & Croy, 2023; Danböck et al., 2021; Krinke et al., 2022; Maslahati et al., 2023). However, most of the included cross-sectional, longitudinal and clinical intervention studies did either not provide this information, or merely stated or implied a 'long' duration since trauma or disorder onset without any further specification (e.g. in the case of childhood trauma investigations in adults). Thus, for any biological measure, it is recommended to specify when it was measured in relation to the trauma exposure and the stage of the disorder of interest.

Over the past decade it has become increasingly clear that the biological mechanisms underlying trauma-related disorders are influenced by sex and sex-related factors, as well as gender (minority)-related factors (Langeland & Olff, 2024). Although the EJPT was the first journal to adopt a gender policy (Olff, 2016) most of the included articles did not explicitly address the potential impact of these factors in their analytic approach. Most articles combined women and men into one sample without investigating potentially differential biological effects of sex and/or gender. Others investigated only women or men but did not take sex and/or gender (minority)-related factors into account. Less than a handful of papers in this special issue purposefully investigated sex and gender-differential effects (Gultig et al., 2023; Schenker et al., 2023) or the impact of sex hormone-related factors (Krinke et al., 2022; Maslahati et al., 2023), although many of the included articles noted this as a relevant limitation in their discussion. Hence, it is recommended that future biological trauma studies further investigate sex and/or gender effects.

4. Conclusions and future research

The large number of articles included in this special issue shows biological research into trauma and trauma-related disorders is thriving. Collectively, the included articles showcase that significant steps are taken to advance our understanding of the biological mechanisms underlying the development of and susceptibility to trauma-related disorders, as well the recovery from these disorders. Our hope is that this increasing knowledge base will eventually

facilitate the development of more effective preventive, early and curative intervention strategies, as well as more effective personalised interventions and timely allocation towards these interventions. This is highly pertinent, as trauma-related disorders remain a significant public health problem given the ongoing high lifetime prevalence of these disorders (Kessler et al., 2017), and the fact that evidence-based treatments are not sufficiently effective for a considerable part of treatment-receiving patients (Semmlinger et al., 2024). In the preceding section we outlined several important recommendations should be taken into account in future biological research into trauma and trauma-related disorders: trauma exposure characteristics; sex and/or gender (minority)-related factors; developmental/lifespan and transgenerational aspects; and the temporal dynamics of biological processes in relation to the stage of the disorder under investigation. We therefore recommend that future biological studies into trauma-related disorders carefully consider which study designs and statistical and computational methodological approaches are appropriate to truly disentangle the effects of these factors on the biological mechanisms behind adverse mental health outcomes of trauma (see Box 1).

Box 1. Recommendations for future biological research on trauma-related disorders

Study design

- Prioritise **prospective, longitudinal, experimental**, and **clinical RCT designs** to better capture temporal dynamics and causal mechanisms.
- Carefully select **appropriate statistical and computational methods** to disentangle complex interactions between biological, psychological, and contextual factors.

Sample diversity

- Include **more diverse and underrepresented populations** across:
  - **Geographic regions**
  - **Developmental stages/lifespan**
  - **Transgenerational effects**

Trauma exposure characteristics

- Clearly specify and incorporate:
  - **Type of trauma**
  - **Trauma history**
  - **Developmental timing of trauma exposure**
- Consider broader definitions of trauma, including **chronic and insidious stressors.**

Temporal dynamics

- Report **when biological measures were taken** relative to:
  - Trauma exposure
  - Stage of disorder (onset, maintenance, recovery)

Sex and gender Considerations

- Explicitly analyse **sex- and gender-related factors**
- Avoid aggregating sexes/genders without testing for differential effects.

Disclosure statement

No potential conflict of interest was reported by the authors.

ORCID

Mirjam van Zuiden http://orcid.org/0000-0002-1225-2702
Miranda Olff http://orcid.org/0000-0003-1016-9515

References

Ansbjerg, M. B., Sandahl, H., Baandrup, L., Jennum, P., & Carlsson, J. (2023). Sleep impairments in refugees diagnosed with post-traumatic stress disorder: A polysomnographic and self-report study. *European Journal of Psychotraumatology*, *14*(1), https://doi.org/10.1080/20008066.2023.2185943

Askovic, M., Soh, N., Elhindi, J., & Harris, A. W. F. (2023). Neurofeedback for post-traumatic stress disorder: Systematic review and meta-analysis of clinical and neurophysiological outcomes. *European Journal of Psychotraumatology*, *14*(2), Taylor and Francis Ltd. https://doi.org/10.1080/20008066.2023.2257435

Baranyi, A., Enko, D., von Lewinski, D., Rothenhäusler, H. B., Amouzadeh-Ghadikolai, O., Harpf, H., Harpf, L., Traninger, H., Obermayer-Pietsch, B., Schweinzer, M., Braun, C. K., & Meinitzer, A. (2021). Assessment of trimethylamine N-oxide (TMAO) as a potential biomarker of severe stress in patients vulnerable to posttraumatic stress disorder (PTSD) after acute myocardial infarction. *European Journal of Psychotraumatology*, *12*(1), https://doi.org/10.1080/20008198.2021.1920201

Beutler, S., Mertens, Y. L., Ladner, L., Schellong, J., Croy, I., & Daniels, J. K. (2022). Trauma-related dissociation and the autonomic nervous system: A systematic literature review of psychophysiological correlates of dissociative experiencing in PTSD patients. *European Journal of Psychotraumatology*, *13*(2). https://doi.org/10.1080/20008066.2022.2132599

Beutler, S., & Croy, I. (2023). Psychophysiological reactions during the trauma-film paradigm and their predictive value for intrusions. *European Journal of Psychotraumatology*, *14*(2), https://doi.org/10.1080/20008066.2023.2281753

Brewin, C. R., Atwoli, L., Bisson, J. I., Galea, S., Koenen, K., & Lewis-Fernández, R. (2025). Post-traumatic stress disorder: Evolving conceptualization and evidence, and future research directions. *World Psychiatry*, *24*(1), 52–80. https://doi.org/10.1002/wps.21269

Burback, L., Brémault-Phillips, S., Nijdam, M. J., McFarlane, A., & Vermetten, E. (2024). Treatment of posttraumatic stress disorder: A state-of-the-art review. *Current Neuropharmacology*, *22*(4), 557–635. https://doi.org/10.2174/1570159X21666230428091433

Bussone, S., Pesca, C., Casetti, V., Croce Nanni, R., Ottaviani, C., Troisi, A., & Carola, V. (2023). The long-term impact of early adverse experience on adaptive functioning: A pilot study integrating measures of mental status, nonverbal communication, and heart rate variability. *European Journal of Psychotraumatology*, *14*(1), https://doi.org/10.1080/20008066.2023.2181766

Campbell, K. A. (2022). The neurobiology of childhood trauma, from early physical pain onwards: As relevant as ever in today's fractured world. *European Journal of Psychotraumatology*, *13*(2), Taylor and Francis Ltd. https://doi.org/10.1080/20008066.2022.2131969

Cusack, S. E., Aliev, F., Bustamante, D., Dick, D. M., & Amstadter, A. B. (2023). A statistical genetic investigation of psychiatric resilience. *European Journal of Psychotraumatology*, *14*(1), https://doi.org/10.1080/20008066.2023.2178762

Danböck, S. K., Rattel, J. A., Franke, L. K., Liedlgruber, M., Miedl, S. F., & Wilhelm, F. H. (2021). Peritraumatic dissociation revisited: Associations with autonomic activation, facial movements, staring, and intrusion formation. *European Journal of Psychotraumatology*, *12*(1), https://doi.org/10.1080/20008198.2021.1991609

dos Santos Oliveira, N. C., Katrinli, S., de Assis, S. G., Smith, A. K., & Serpeloni, F. (2023). Community and domestic violence are associated with DNA methylation GrimAge acceleration and heart rate variability in adolescents. *European Journal of Psychotraumatology*, *14*(2), https://doi.org/10.1080/20008066.2023.2202054

Elam, K. K., Johnson, S. L., Ruof, A., Eisenberg, D. T. A., Rej, P. H., Sandler, I., & Wolchik, S. (2022). Examining the influence of adversity, family contexts, and a family-based intervention on parent and child telomere length. *European Journal of Psychotraumatology*, *13*(1), https://doi.org/10.1080/20008198.2022.2088935

Engel, S., Laufer, S., Klusmann, H., Schulze, L., Schumacher, S., & Knaevelsrud, C. (2023). Cortisol response to traumatic stress to predict PTSD symptom development – a systematic review and meta-analysis of experimental studies. *European Journal of Psychotraumatology*, *14*(2), Taylor and Francis Ltd. https://doi.org/10.1080/20008066.2023.2225153

Engel, S., Schumacher, S., Niemeyer, H., Kuester, A., Burchert, S., Klusmann, H., Rau, H., Willmund, G. D., & Knaevelsrud, C. (2021). Associations between oxytocin and vasopressin concentrations, traumatic event exposure and posttraumatic stress disorder symptoms: Group comparisons, correlations, and courses during an internet-based cognitive-behavioural treatment. *European Journal of Psychotraumatology*, *12*(1), https://doi.org/10.1080/20008198.2021.1886499

Fine, N. B., Ben-Zion, Z., Biran, I., & Hendler, T. (2023). Neuroscientific account of guilt- and shame-driven PTSD phenotypes. *European Journal of Psychotraumatology*, *14*(2), https://doi.org/10.1080/20008066.2023.2202060

Glazebrook, A. J., Shakespeare-Finch, J., Andrew, B., & van der Meer, J. (2023). Posttraumatic growth EEG neuromarkers: Translational neural comparisons with resilience and PTSD in trauma-exposed healthy adults. *European Journal of Psychotraumatology*, *14*(2), https://doi.org/10.1080/20008066.2023.2272477

Grasser, L. R., Saad, B., Bazzi, C., Wanna, C., Abu Suhaiban, H., Mammo, D., Jovanovic, T., & Javanbakht, A. (2022). Skin conductance response to trauma interview as a candidate biomarker of trauma and related psychopathology in youth resettled as refugees. *European Journal of Psychotraumatology*, *13*(1), https://doi.org/10.1080/20008198.2022.2083375

Gultig, K. D., de Rooij, S. R., Hilberdink, C. E., Olff, M., Roseboom, T. J., & van Zuiden, M. (2023). Effects of prenatal exposure to the 1944–45 Dutch famine and glucocorticoid receptor polymorphisms on later life PTSD susceptibility. *European Journal of Psychotraumatology*, *14*(2), https://doi.org/10.1080/20008066.2023.2219075

He, J., Zhong, X., Cheng, C., Dong, D., Zhang, B., Wang, X., & Yao, S. (2023). Characteristics of white matter structural connectivity in healthy adults with childhood maltreatment. *European Journal of Psychotraumatology*, *14*(1), https://doi.org/10.1080/20008066.2023.2179278

Hilberdink, C. E., van Zuiden, M., Schrantee, A., Korosi, A., Kaiser, A., Zhutovsky, P., Ginty, A. T., Ensink, J. B. M., Lindauer, R. J. L., Vrijkotte, T. G. M., & de Rooij, S. R. (2021). Dysregulated functional brain connectivity in response to acute social-evaluative stress in adolescents with PTSD symptoms. *European Journal of Psychotraumatology*, *12*(1), https://doi.org/10.1080/20008198.2021.1880727

Jin, X., Xu, B., Xu, R., Yin, X., Yan, S., Zhang, Y., & Jin, H. (2023). The influence of childhood emotional neglect experience on brain dynamic functional connectivity in young adults. *European Journal of Psychotraumatology*, *14*(2), https://doi.org/10.1080/20008066.2023.2258723

Kessler, R. C., Aguilar-Gaxiola, S., Alonso, J., Benjet, C., Bromet, E. J., Cardoso, G., Degenhardt, L., de Girolamo, G., Dinolova, R. V., Ferry, F., Florescu, S., Gureje, O., Haro, J. M., Huang, Y., Karam, E. G., Kawakami, N., Lee, S., Lepine, J.-P., Levinson, D., … Koenen, K. C. (2017). Trauma and PTSD in the WHO world mental health surveys. *European Journal of Psychotraumatology*, *8*(sup5), https://doi.org/10.1080/20008198.2017.1353383

Krause-Utz, A., Walther, J. C., Kyrgiou, A. I., Hoogenboom, W., Alampanou, M., Bohus, M., Schmahl, C., & Lis, S. (2022). Severity of childhood maltreatment predicts reaction times and heart rate variability during an emotional working memory task in borderline personality disorder. *European Journal of Psychotraumatology*, *13*(1), https://doi.org/10.1080/20008198.2022.2093037

Krinke, E., Held, U., Steigmiller, K., Felmingham, K., & Kleim, B. (2022). Sex hormones and cortisol during experimental trauma memory consolidation: Prospective association with intrusive memories. *European Journal of Psychotraumatology*, *13*(1), https://doi.org/10.1080/20008198.2022.2040818

Kurkinen, K., Kärkkäinen, O., Lehto, S. M., Luoma, I., Kraav, S. L., Kivimäki, P., Nieminen, A. I., Sarnola, K., Therman, S., & Tolmunen, T. (2023). The associations between metabolic profiles and sexual and physical abuse in depressed adolescent psychiatric outpatients: An exploratory pilot study. *European Journal of Psychotraumatology*, *14*(1), https://doi.org/10.1080/20008066.2023.2191396

Lan, Q., Ge, J., Dai, H., Lu, J., Wu, L., Liu, B., Zhang, L., Lu, G., Qi, R., Cao, Z., & Luo, Y. (2023). Longitudinal changes in brain structure and their relationship with subclinical psychiatric symptoms in parents who lost their only child in China. *European Journal of Psychotraumatology*, *14*(2), https://doi.org/10.1080/20008066.2023.2216624

Langeland, W., & Olff, M. (2024). Sex and gender in psychotrauma research. *European Journal of Psychotraumatology*, *15*(1), Taylor and Francis Ltd. https://doi.org/10.1080/20008066.2024.2358702

Lee, S., Park, J. T., Bang, M., An, S. K., Namkoong, K., Park, H. Y., & Lee, E. (2022). Theory of mind and hair cortisol in healthy young adults: The moderating effects of childhood trauma. *European Journal of Psychotraumatology*, *13*(2), https://doi.org/10.1080/20008066.2022.2116826

Liddell, B. J., Das, P., Malhi, G. S., Nickerson, A., Felmingham, K. L., Askovic, M., Aroche, J., Coello, M., Cheung, J., Den, M., Outhred, T., & Bryant, R. A. (2023). Refugee visa insecurity disrupts the brain's default mode network. *European Journal of Psychotraumatology*, *14*(2), https://doi.org/10.1080/20008066.2023.2213595

Maffioletti, E., Bocchio-Chiavetto, L., Perusi, G., Carvalho Silva, R., Sacco, C., Bazzanella, R., Zampieri, E., Bortolomasi, M., Gennarelli, M., & Minelli, A. (2021).

Inflammation-related microRNAs are involved in stressful life events exposure and in trauma-focused psychotherapy in treatment-resistant depressed patients. *European Journal of Psychotraumatology*, *12*(1), https://doi.org/10.1080/20008198.2021.1987655

Magnin, C., Poulet, E., Fanton, L., Vignaud, P., & Brunelin, J. (2023). A non-randomized pilot trial of the use of prazosin in the prevention of transition from acute stress disorder to post-traumatic stress disorder. *European Journal of Psychotraumatology*, *14*(2), https://doi.org/10.1080/20008066.2023.2251250

Manthey, A., Sierk, A., Brakemeier, E. L., Walter, H., & Daniels, J. K. (2021). Does trauma-focused psychotherapy change the brain? A systematic review of neural correlates of therapeutic gains in PTSD. *European Journal of Psychotraumatology*, *12*(1), Taylor and Francis Ltd. https://doi.org/10.1080/20008198.2021.1929025

Marx, B. P., Hall-Clark, B., Friedman, M. J., Holtzheimer, P., & Schnurr, P. P. (2024). The PTSD criterion a debate: A brief history, current status, and recommendations for moving forward. *Journal of Traumatic Stress*, *37*(1), 5–15. John Wiley and Sons Inc. https://doi.org/10.1002/jts.23007

Maslahati, T., Schultebraucks, K., Galve Gómez, M., Hellmann-Regen, J., Otte, C., Wingenfeld, K., & Roepke, S. (2023). Effects of oral contraceptives on intrusive memories: A secondary analysis of two studies using the trauma film paradigm in healthy women. *European Journal of Psychotraumatology*, *14*(2), https://doi.org/10.1080/20008066.2023.2282003

Mattonet, K., Scharpf, F., Block, K., Kumsta, R., & Hecker, T. (2023). No association between war-related trauma or PTSD symptom severity and epigenome-wide DNA methylation in Burundian refugees. *European Journal of Psychotraumatology*, *14*(2), https://doi.org/10.1080/20008066.2023.2228155

Nicholson, A. A., Siegel, M., Wolf, J., Narikuzhy, S., Roth, S. L., Hatchard, T., Lanius, R. A., Schneider, M., Lloyd, C. S., McKinnon, M. C., Heber, A., Smith, P., & Lueger-Schuster, B. (2022). A systematic review of the neural correlates of sexual minority stress: Towards an intersectional minority mosaic framework with implications for a future research agenda. *European Journal of Psychotraumatology*, *13*(1), Taylor and Francis Ltd. https://doi.org/10.1080/20008198.2021.2002572

Noble, N. C., Merker, J. B., Webber, T. K., Ressler, K. J., & Seligowski, A. V. (2023). PTSD and depression severity are associated with cardiovascular disease symptoms in trauma-exposed women. *European Journal of Psychotraumatology*, *14*(2), https://doi.org/10.1080/20008066.2023.2234810

Olff, M. (2016). Five years of *European Journal of Psychotraumatology*. *European Journal of Psychotraumatology*, *7*(1), Co-Action Publishing. https://doi.org/10.3402/ejpt.v7.31350

Powers, A., Hinojosa, C. A., Stevens, J. S., Harvey, B., Pas, P., Rothbaum, B. O., Ressler, K. J., Jovanovic, T., & van Rooij, S. J. H. (2022). Right inferior frontal gyrus and ventromedial prefrontal activation during response inhibition is implicated in the development of PTSD symptoms. *European Journal of Psychotraumatology*, *13*(1), https://doi.org/10.1080/20008198.2022.2059993

Rosada, C., Bauer, M., Golde, S., Metz, S., Roepke, S., Otte, C., Wolf, O. T., Buss, C., & Wingenfeld, K. (2021). Association between childhood trauma and brain anatomy in women with post-traumatic stress disorder, women with borderline personality disorder, and healthy women. *European Journal of Psychotraumatology*, *12*(1), https://doi.org/10.1080/20008198.2021.1959706

Roullet, P., Taïb, S., Thalamas, C., Vaiva, G., El Hage, W., Yrondi, A., & Birmes, P. (2022). Efficacy of traumatic memory reactivation with or without propranolol in PTSD with high dissociative experiences. *European Journal of Psychotraumatology*, *13*(2), https://doi.org/10.1080/20008066.2022.2151098

Salvat-Pujol, N., Labad, J., Urretavizcaya, M., de Arriba-Arnau, A., Segalàs, C., Real, E., Ferrer, A., Crespo, J. M., Jiménez-Murcia, S., Soriano-Mas, C., Menchón, J. M., & Soria, V. (2021). Childhood maltreatment interacts with hypothalamic-pituitary-adrenal axis negative feedback and major depression: Effects on cognitive performance. *European Journal of Psychotraumatology*, *12*(1), https://doi.org/10.1080/20008198.2020.1857955

Schellhaas, S., Schmahl, C., & Bublatzky, F. (2022). Social threat and safety learning in individuals with adverse childhood experiences: Electrocortical evidence on face processing, recognition, and working memory. *European Journal of Psychotraumatology*, *13*(2), https://doi.org/10.1080/20008066.2022.2135195

Schenker, M. T., Theoswin, P. M., Qian, H., Jordan, A. S., Nicholas, C. L., & Felmingham, K. L. (2023). Sleep and day-to-day PTSD symptom variability: An ecological momentary assessment and actigraphy monitored study in trauma-exposed young adults. *European Journal of Psychotraumatology*, *14*(2), https://doi.org/10.1080/20008066.2023.2202058

Schmitz, M., Müller, L. E., Seitz, K. I., Schulz, A., Steinmann, S., Herpertz, S. C., & Bertsch, K. (2021). Heartbeat evoked potentials in patients with post-traumatic stress disorder: An unaltered neurobiological regulation system? *European Journal of Psychotraumatology*, *12*(1), https://doi.org/10.1080/20008198.2021.1987686

Semmlinger, V., Leithner, C., Klöck, L. M., Ranftl, L., Ehring, T., & Schreckenbach, M. (2024). Prevalence and predictors of nonresponse to psychological treatment for PTSD: A meta-analysis. *Depression and Anxiety, 2024*, https://doi.org/10.1155/2024/9899034

Shalev, A., Cho, D., & Marmar, C. R. (2024). Neurobiology and treatment of posttraumatic stress disorder. *American Journal of Psychiatry*, *181*(8), 705–719. https://doi.org/10.1176/appi.ajp.20240536

Sokołowski, A., Kowalski, J., & Dragan, M. (2022). Neural functional connectivity during rumination in individuals with adverse childhood experiences. *European Journal of Psychotraumatology*, *13*(1), https://doi.org/10.1080/20008198.2022.2057700

Susanty, E., Sijbrandij, M., van Dijk, W., Srisayekti, W., de Vries, R., & Huizink, A. C. (2022). The effects of psychological interventions on neurocognitive functioning in posttraumatic stress disorder: A systematic review. *European Journal of Psychotraumatology*, *13*(1), Taylor and Francis Ltd. https://doi.org/10.1080/20008198.2022.2071527

Tamman, A. J. F., Jiang, L., Averill, C. L., Mason, G. F., Averill, L. A., & Abdallah, C. G. (2023). Biological embedding of early trauma: The role of higher prefrontal synaptic strength. *European Journal of Psychotraumatology*, *14*(2), https://doi.org/10.1080/20008066.2023.2246338

van Sleeuwen, C., van Zuiden, M., Koch, S. B. J., Frijling, J. L., Veltman, D. J., Olff, M., & Nawijn, L. (2023). How does it feel? An exploration of neurobiological and clinical correlates of alexithymia in trauma-exposed police-officers with and without PTSD. *European Journal of*

Psychotraumatology, *14*(2), https://doi.org/10.1080/20008066.2023.2281187

Vicario, C. M., Martino, G., Lucifora, C., & Felmingham, K. (2022). Preliminary evidence on the neural correlates of timing deficit in post-traumatic stress disorder. *European Journal of Psychotraumatology*, *13*(1), Taylor and Francis Ltd. https://doi.org/10.1080/20008198.2021.2008151

Vuong, E., Hemmings, S. M., Mhlongo, S., Chirwa, E., Lombard, C., Peer, N., Abrahams, N., & Seedat, S. (2022). Adiponectin gene polymorphisms and posttraumatic stress disorder symptoms among female rape survivors: An exploratory study. *European Journal of Psychotraumatology*, *13*(2), https://doi.org/10.1080/20008066.2022.2107820

Wadji, D. L., Gaillard, C., Ketcha Wanda, G. J. M., Wicky, C., Morina, N., & Martin-Soelch, C. (2021). HPA-axis activity and the moderating effect of self-esteem in the context of intimate partner violence in Cameroon. *European Journal of Psychotraumatology*, *12*(1), https://doi.org/10.1080/20008198.2021.1930897

Wermuth, K., Ülsmann, D., Borngräber, J., Gallinat, J., Schulte-Herbrüggen, O., & Kühn, S. (2021). Structural signature of trauma: White matter volume in right inferior frontal gyrus is positively associated with use of expressive suppression in recently traumatized individuals. *European Journal of Psychotraumatology*, *12*(1), https://doi.org/10.1080/20008198.2020.1837512

Williams, K. C., Nto, N. J., van Vuren, E. J., Sallie, F. N., Molebatsi, K., Kroneberg, K. S., Roomaney, A. A., Salie, M., & Womersley, J. S. (2024). Early biological and psychosocial factors associated with PTSD onset and persistence in youth. *European Journal of Psychotraumatology*, *15*(1), Taylor and Francis Ltd. https://doi.org/10.1080/20008066.2024.2432160

Xiao, Q., Fu, Y., Yi, X., Ding, J., Han, Z., Zhang, Z., Tan, Z., Wang, J., Wu, Z., Pi, J., & Chen, B. T. (2023). Altered cortical thickness and emotional dysregulation in adolescents with borderline personality disorder. *European Journal of Psychotraumatology*, *14*(1), https://doi.org/10.1080/20008066.2022.2163768

Yrondi, A., Colineaux, H., Claudet, I., Sales de Gauzy, J., Huo, S., Taib, S., Bui, E., & Birmes, P. (2022). Prevalence and prediction of PTSD and depression in mothers of children surviving a motor vehicle crash. *European Journal of Psychotraumatology*, *13*(2), https://doi.org/10.1080/20008066.2022.2121014

Zhao, H., Dong, D., Sun, X., Cheng, C., Xiong, G., Wang, X., & Yao, S. (2021). Intrinsic brain network alterations in non-clinical adults with a history of childhood trauma. *European Journal of Psychotraumatology*, *12*(1), https://doi.org/10.1080/20008198.2021.1975951